

# Clara Dehman

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*Department of Physics, University of Alicante, 03690 Alicante, Spain*

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## Education

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### Ph.D. in Physics

Nov 2019 — Nov 2023

*Autonomous University of Barcelona, Spain*

- *Thesis Title:* Unveiling the Physics of Neutron Stars: A 3D expedition into MAGNETO-Thermal evolution in Isolated Neutron Stars with “MATINS”
- *Honors:* Cum Laude Distinction · International Mention · IEEC Best Thesis Award · SEA Thesis Award (Special Mention)

### International Joint M.Sc. in Theoretical Nuclear Physics

Oct 2017 — Sep 2019

*Erasmus Mundus NucPhys Program, European Union*

Degrees Awarded:

- Laurea Magistrale in Fisica, *University of Padova, Italy* — 110/110 + Cum Laude
- Master de Sciences, Technologies, Santé, Mention Physique, *University of Caen Normandy, France*
- Máster Universitario Erasmus Mundus en Física Nuclear, *Spanish Consortium Universities* — Matrícula de honor

### B.Sc. in Physics

Oct 2014 — Aug 2017

*Lebanese University, Lebanon* — Ranked 1st in Physics Department

## Research Positions

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### Juan de la Cierva Postdoctoral Fellow

Mar 2024 — present

*University of Alicante, Spain*

- *Project:* Investigate the chiral magnetic effect, magnetic helicity, origin of magnetar magnetic fields, long-period transients, and sub-grid scaling.
- *Advisor:* Jose A. Pons

### Transitional Postdoctoral Researcher

Nov 2023 — Feb 2024

*Institute of Space Sciences (ICE-CSIC), Barcelona, Spain*

- *Project:* Modeled 3D neutron star magnetic field evolution and benchmarked results against X-ray observations.
- *Advisor:* Nanda Rea

### Visiting Ph.D. Fellow

Apr — Dec 2023

*Nordita, Stockholm, Sweden*

- *Project:* Studied the role of magnetic helicity in the inverse cascade of neutron star magnetic fields.
- *Advisor:* Axel Brandenburg

**Doctoral Researcher****Nov 2019 — Nov 2023***Institute of Space Sciences, Barcelona, Spain*

- *Project:* Developed MATINS, a 3D code for magneto-thermal evolution of isolated neutron stars; applied it to cooling ( $\pm$  hyperons), magnetar outbursts, and interior–magnetosphere coupling using physics-informed neural networks.
- *Advisors:* Nanda Rea, Daniele Viganò, Jose A. Pons

**Graduate Researcher****Feb — Sep 2019***University of Barcelona, Spain**Grade: 10/10 + Matrícula de Honor*

- *Project:* Computed finite-temperature equations of state using the BCPM energy density functional for post neutron star mergers and late-stage proto-neutron stars.
- *Advisors:* Artur Polls, Xavier Viñas, Mario Centelles

**Intern****Sep 2018 — Jan 2019***Laboratory of Corpuscular Physics, Caen, France*

- *Project:* Developed metamodeling approaches for nuclear equations of state to interpret experimental data, focusing on the neutron star mass–radius relation and tidal deformability.
- *Advisors:* Francesca Gulminelli

**Intern****Mar — Jun 2018***National Institute for Nuclear Physics, Legnaro Laboratories, Italy*

- *Project:* Measured fusion cross sections of a  $^{28}\text{Si}$  beam on a  $^{100}\text{Mo}$  target at various energies to probe many-body quantum tunneling.
- *Advisors:* Giovanna Montagnoli, Alberto Stefanini

**Fellowships & Research Fundings****COMPETITIVE FELLOWSHIPS & SCHOLARSHIPS AS PI***Total secured: €120,000, excluding €92,470 APOSTD Fellowship (declined)*

- **Feb 2025 – Jan 2027:** *Juan de la Cierva Postdoctoral Fellowship*, Spanish Ministry of Science and Innovation — nationally competitive (10% success rate) — €72,000
- **Oct 2024 – Sep 2026:** *APOSTD Postdoctoral Fellowship*, Government of Valencia, Spain — €92,470 (declined in favor of the more prestigious Juan de la Cierva Fellowship).
- **Apr 2023 – Dec 2023:** *Nordita Visiting PhD Fellowship*, Nordita, Sweden — 80,000 kr
- **Oct 2017 – Sep 2019:** *EACEA Scholarship*, Erasmus Mundus Joint MSc Program, European Union — €37,000
- **Sep 2012 – Jun 2014:** *English Micro-Scholarship Program*, American Embassy, Lebanon — \$2,000

**COLLABORATIVE RESEARCH GRANTS AS CONTRIBUTOR***Contributed to research grants totaling €2,944,000*

- **Mar 2024 – Dec 2026:** *BLADES (Birth, Life, And Death of Extreme Sources)*, CIPROM 2022/13 Grant, Government of Valencia, Spain — PI: Jose A. Pons, €584,000
- **Mar 2024 – Dec 2026:** *Relativistic Computational Astrophysics*, ASFAE/2022/26 Grant, Government of Valencia, Spain — PI: Miguel A. Aloy, €300,000

- **Apr 2022 – Dec 2024:** *SGR-Cat 2021 (01269) Grant*, Government of Catalonia, Spain — PI: Vanessa Graber, €60,000
- **Nov 2019 – Feb 2024:** *Magnesia (The magnetar census and impact of highly magnetic neutron stars on the explosive and transient Universe) ERC Consolidator Grant*, European Research Council — PI: Nanda Rea, €2,000,000

## Awards & Certificates

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- **Dec 2024:** *IEEC Best Doctoral Thesis Award 2024* — \$1,200 cash + \$600 research support
- **May 2024:** *Special Mention, SEA Thesis Award 2024* — for outstanding research quality in a highly competitive category
- **Mar 2019:** *Certificate in Data Analysis and Machine Learning*, Michigan State University (USA) and University of Oslo (Norway)
- **Apr 2017:** *ERASMUS+ Joint Master Program (NucPhys)* — selected among top 20% of international applicants (CV and Interview)

## Invited Talks & Seminars (16 Total)

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- **Mar 2026:** *Strong Gravity Seminar*, Perimeter Institute, Waterloo, Canada
- **Mar 2026:** *Seminar*, Canadian Institute for Theoretical Astrophysics, Canada
- **Jun 2025:** *Talk*, Fòrum IEEC 2025, Casa Convalescència (UAB), Barcelona, Spain
- **May 2025:** *Talk*, Extreme Physics of Neutron Star Interiors, Princeton University, USA
- **May 2025:** *Review Talk*, Workshop on Modern Equation of State and Spectroscopy in Neutron Stars, Alcalá University, Spain
- **Feb 2025:** *Talk*, Breaking New Ground in Supernova Physics, Fukuoka University, Japan
- **Dec 2024:** *Review Talk*, IReNA-INT Joint Workshop on Thermal and Magnetic Evolution, University of Washington, USA
- **Sep 2024:** *Talk*, Frontier Workshop in Astrophysics IV, Palermo, Italy
- **Jun 2023:** *Astrophysics Seminar*, University of Valencia, Spain
- **May 2023:** *Astrophysics Seminar*, Nordita, Sweden [\[Link\]](#)
- **Jun 2022:** *Astrocoffee Seminar*, Goethe University, Germany [\[Link\]](#)
- **May 2022:** *Seminar*, NucPhys Master Program, GANIL, France
- **Dec 2021:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain [\[Link\]](#)
- **Nov 2021:** *Astrophysics Seminar*, University of Alicante, Spain
- **Dec 2020:** *Talk*, Stars & Compact Objects Meeting, Flatiron Institute, New York, USA
- **Mar 2019:** *Seminar*, Hadronic, Nuclear & Atomic Physics Group, University of Barcelona, Spain

## Contributed Talks & Seminars (11 Total)

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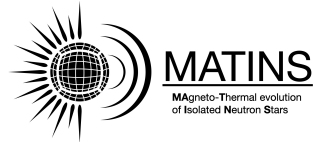
- **Jun 2025:** *Talk*, EAS Annual Meeting, University College Cork, Ireland
- **Jul 2024:** *Talk*, 10<sup>th</sup> International Conference on Quarks and Nuclear Physics, University of Barcelona, Spain
- **Jun 2024:** *Talk*, XMM-Newton Workshop 2024, Madrid, Spain
- **May 2024:** *Special Seminar*, Institute of Space Sciences, Barcelona, Spain

- **Apr 2024:** *Astrophysics Seminar*, University of Alicante, Spain
- **May 2022:** *Talk*, Pharos Conference, University of La Sapienza, Rome, Italy
- **Mar 2022:** *Seminar*, Institute of Space Sciences, Barcelona, Spain [[Link](#)]
- **Nov 2021:** *Talk*, IAU Symposium 363 (online)
- **Jun 2021:** *Talk*, EAS Annual Meeting (online)
- **Nov 2020:** *Seminar*, Institute of Space Sciences, Barcelona, Spain [[Link](#)]
- **Mar 2020:** *Talk*, PHAROS Conference, Patras, Greece (canceled due to COVID-19)

## Software Development & Open Access Tools

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**Since Sep 2020:** Lead developer of MATINS — a 3D code for the MAgneto-Thermal evolution in Isolated Neutron Stars — [MATINS webpage](#)



- *Open access:* [github.com/ice-csic-astroexotic/MATINS](https://github.com/ice-csic-astroexotic/MATINS)
- *Lead developers:* **C. Dehman**, D. Viganò, S. Ascenzi
- *Implementation:* Written in Fortran 90 with OpenMP parallelization, optimized for Intel compilers (better performance than GNU).

## Supervision

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**PhD Students** — *Co-supervisor; University of Alicante, Spain*

- Carlos Rábano Alcoba — Subgrid-scale modeling (since Jan 2026)
- Joan Llorens Ripoll — Ambipolar diffusion in neutron star cores (since Mar 2025)

**MSc Students** — *Co-mentor; Institute of Space Sciences, Spain*

- Marco Ermolli — Impact of microphysics on neutron star cooling (Jan – Jul 2021)

**BSc Students** — *Main supervisor; University of Alicante, Spain*

- Ahlam Makboul — Glitch timing and magnitude of PSR J0537–6910: probing crust–magnetic field interactions (since Oct 2025)

## Teaching (112 Hours Total)

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**Instructor — Introduction to Modeling in Physics**

*University of Alicante, Spain*

**Editions 2024 & 2025**

45 hr — 95 BSc students

**Instructor — Nuclear and Particle Physics**

*University of Alicante, Spain*

**Editions 2024 & 2025**

54 hr — 112 BSc students

**Lecturer — Stellar Astrophysics**

*University of Alicante, Spain*

**Jan — Mar 2025**

12 hr — 14 BSc students

**Invited Lecturer — Research in Computational Astrophysics**

*University of Alicante, Spain*

**May 2025**

1 hr — 10 MSc students

## Scientific Community Service, Outreach & Visibility

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### SCIENTIFIC MEETING ORGANIZATION

*SOC: Scientific Organizing Committee; LOC: Local Organizing Committee*

- **Autumn 2026:** *SOC & LOC*, 22nd Pencil Code User Meeting, University of Alicante, Spain
- **Autumn 2026:** *SOC*, Modeling neutron star magnetic fields, Lorentz Center, Netherlands
- **Nov 2022:** *LOC*, Exploring the Hot and Energetic Universe: the 3rd scientific conference dedicated to the Athena X-ray observatory, Barcelona, Spain [[Link](#)]
- **May 2022:** *LOC*, PHAROS Final Conference 2022: The multi-messenger physics and astrophysics of neutron stars, University of La Sapienza, Rome, Italy [[Link](#)]

### REVIEWING & EVALUATION ACTIVITIES

- **Dec 2025:** *PhD committee member* – Juan A. Gil Granados, Univ. of Barcelona, Spain
- **Since Jun 2025:** *Expert evaluator for PhD proposals*, Valencian Agency for Assessment and Prospective, Spain
- **Since 2022:** *Reviewer* for Nature Astronomy; Journal of Physics G; Monthly Notices of the Royal Astronomical Society; Frontiers in Astronomy and Space Sciences; Classical and Quantum Gravity

### OUTREACH ACTIVITIES

- **Nov 2022:** *Astronomical Observation* — Contributed to organizing an astronomical observation (Autonomous University of Barcelona, Spain; [[Link](#)])
- **Sep 2022:** *Research Night* — Observation night for school children (Igualada, Spain)
- **Feb 2022:** *Outreach talk* — International Day of Girls and Women in Science (ICE-CSIC, Spain; [[Link](#)])
- **2021–2023:** *Magnet ICE Outreach* — Collaborated with teachers at Gabriel Castellà i Raich in creating innovative science projects to promote student integration in a school facing segregation issues (Igualada, Spain).

### PRESS RELEASES (14 TOTAL)

- **Jul 2024:** *Descubren tres estrellas de neutrones jóvenes inusualmente frías para su edad*, [europapress](#)
- **Jul 2024:** *Investigadores de la UA detectan estrellas de neutrones inusualmente frías*, [TODOAlicante](#)
- **Jul 2024:** *Investigadores de la UA participan en el descubrimiento de tres estrellas de neutrones demasiado frías para su edad*, [Actualidad Universitaria](#)
- **Jun 2024:** *NASA's Chandra Peers Into Densest and Weirdest Stars*, [NASA](#)
- **Jun 2024:** *Too young to be so cool: lessons from three neutron stars*, [ESA](#)
- **Jun 2024:** *Un equip d'astrònoms de l'IEEC descobreix tres estrelles de neutrons massa fredes per a la seva edat*, [IEEC](#)
- **Jun 2024:** *Troppo giovani per essere così fredde*, [MEDIA INAF](#)
- **Jun 2024:** *A team of ICE-CSIC astronomers discovers three neutron stars too cold for their age*, [ICE-CSIC](#)
- **Dec 2022:** *Astronomers observe outburst of the young magnetar Swift J1818.0–1607*, [Phys Org](#)

- **Jun 2020:** *NASA discovers youngest ‘cosmic baby’ neutron star*, [Independent](#)
- **Jun 2020:** *Astronomers just discovered the youngest ever ‘baby’ dead star*, [CNET](#)
- **Jun 2020:** *Astronomers find baby pulsar – a rare magnetar – born just 240 years ago*, [Astronomy Now](#)
- **Jun 2020:** *Astronomers discover youngest magnetar ever*, [Sci News](#)
- **Jun 2020:** *Hallan estrella ‘bebé’ que explica origen de las explosiones en el universo*, [La Vanguardia](#)

## INVITED INTERVIEW

- **Jun 2024:** *Invited Interview* at University of Elche, Spain [[Link](#)]
- **Dec 2023:** *Invited Interview* at ICE-CSIC, Barcelona, Spain [[Interview](#); [Newsletter](#)]

## Languages, Memberships & Trainings

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### LEADERSHIP TRAINING

Project Citizen (AMIDEAST & American Embassy, Lebanon, [2014](#)) • Intel Youth Innovation Camp (INJAZ, [2013](#)) • Personal Strategic Planning (IAPTL, [2013](#)) • Leadership Program (AMIDEAST & INJAZ, [2012](#))

### COLLABORATIONS & MEMBERSHIPS

NewAthena Science Community ([since Mar 2025](#)) • Einstein Telescope Collaboration ([since Sep 2022](#)) • AIHUB, Artificial Intelligence Initiative of Spain’s National Research Council ([since Mar 2022](#)) • Pharos European COST Action CA16214 ([2019 – 2022](#))

### LANGUAGES

[Arabic](#) (C2, mother tongue) • [English](#) (C2, fluent; Advanced Cambridge 183/210, TOEFL 805/900) • [French](#) (C2, fluent) • [Spanish](#) (B2, intermediate) • [Italian](#) (B2, intermediate)



## List of Publications

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Curated selection of 21 peer-reviewed articles: 7 as first author, 2 as co-first author (including one in Nature Astronomy), and 1 invited Living Review, accessible via [ADS](#), [arXiv](#), [ORCID](#), and [Google Scholar](#) — **Citations:** 400+ | **h-index:** 13

### SUBMITTED:

**21.** J.A. Pons, **C. Dehman**, D. Viganò: 2025. *Magnetic, thermal and rotational evolution of isolated neutron stars*. Submitted to Living Reviews in Computational Astrophysics, Sep 2025 ([arXiv:2509.06699](#)).

**20.** A. Suvorov, **C. Dehman\***, J.A. Pons: 2025. *Late-blooming magnetars: awakening as ultra-long period objects after a dormant cooling epoch*. Submitted to Nature Commun., May 2025 (Equal contribution by first two authors; corresponding author: C. Dehman). ([arXiv:2505.05373](#)).

### PUBLISHED:

**19.** A. Suvorov, **C. Dehman**, J.A. Pons: 2025. *Revealing the nature of ultra-long period objects with space-based gravitational-wave interferometers*. ApJ 991, 134 ([arXiv:2505.06125](#), [DOI](#)).

**18.** **C. Dehman** and J.A. Pons: 2025. *Magnetar field dynamics shaped by chiral anomalies and helicity*. Phys. Rev. Research 7, 033231 ([arXiv:2505.06196](#), [DOI](#)).

**17.** **C. Dehman** and A. Brandenburg: 2025. *Reality of inverse cascading in neutron star crusts*. A&A 694, A39 ([arXiv:2408.08819](#), [DOI](#)).

**16.** S. Ascenzi, D. Viganò, **C. Dehman** et al.: 2024. *3D code for MAGneto-Thermal evolution in Isolated Neutron Stars, MATINS: thermal evolution and light curves*. MNRAS 533, 201 ([arXiv:2401.15711](#), [DOI](#)).

**15.** **C. Dehman**, M. Centelles, X. Viñas: 2024. *Impact of the hot inner crust on compact stars at finite temperature*. A&A 687, A236 ([arXiv:2401.16957](#), [DOI](#)).

**14.** A. Marino\*, **C. Dehman\***, K. Kovlakas\*, N. Rea\* et al.: 2024. *Constraints on the dense matter equation of state from young and cold isolated neutron stars*. Nature Astron. 8, 1020 ([arXiv:2404.05371](#), [DOI](#)). (Equal contribution by these authors).

**13.** J. Urbán, P. Stefanou, **C. Dehman**, J.A. Pons: 2023. *Modeling force-free neutron star magnetospheres using physics-informed neural networks*. MNRAS 524, 32 ([arXiv:2303.11968](#), [DOI](#)).

**12.** **C. Dehman**, D. Viganò, S. Ascenzi et al.: 2023. *3D evolution of neutron star magnetic-fields from a realistic core-collapse turbulent topology*. MNRAS 523, 5198 ([arXiv:2305.06342](#), [DOI](#)).

**11.** **C. Dehman**, J.A. Pons, D. Viganò, N. Rea: 2023. *How bright can old magnetars be? Assessing the impact of magnetized envelopes and field topology on neutron star cooling*. MNRAS Lett. 520, L42 ([arXiv:2301.02261](#), [DOI](#)).

**10.** A.Y. Ibrahim, A. Borghese, N. Rea et al.: 2023. *Deep X-ray and radio observations of the first outburst of the young magnetar Swift J1818.0-1607*. ApJ 943, 20 ([arXiv:2211.12391](#), [DOI](#)).

**9.** **C. Dehman**, D. Viganò, J.A. Pons, N. Rea: 2023. *3D code for MAGneto-Thermal evolution in Isolated Neutron Stars, MATINS: the magnetic field formalism*. MNRAS 518, 1222 ([arXiv:2209.12920](#), [DOI](#)).

**8.** N. Rea, F. Coti Zelati, **C. Dehman** et al.: 2022. *Constraining the nature of the 18 min periodic radio transient GLEAM-X J162759.5-523504.3 via multi-wavelength observations and magneto-thermal simulations*. ApJ 940, 72 ([arXiv:2210.01903](#), [DOI](#)).

7. F. Anzuini, A. Melatos, **C. Dehman** et al.: 2022. *Thermal luminosity degeneracy of magnetized neutron stars with and without hyperon cores*. MNRAS 515, 3014 ([arXiv:2205.14793](#),[DOI](#)).
6. F. Anzuini, A. Melatos, **C. Dehman** et al.: 2022. *Fast cooling and internal heating in hyperon stars*. MNRAS 509, 2609 ([arXiv:2110.14039](#),[DOI](#)).
5. D. Viganò, A. Garcia-Garcia, J.A. Pons, **C. Dehman** et al.: 2021. *Magneto-thermal evolution of neutron stars with coupled Ohmic, Hall and ambipolar effects via accurate finite-volume simulations*. Comput. Phys. Commun. 265, 108001 ([arXiv:2104.08001](#),[DOI](#)).
4. A.M. Stefanini, G. Montagnoli, M. D'Andrea, M. Giacomini, **C. Dehman** et al.: 2021. *New insights into sub-barrier fusion of  $^{28}\text{Si} + ^{100}\text{Mo}$* . J. of Phys. G 48, 055101 ([ADS](#),[DOI](#)).
3. F. Coti Zelati, A. Borghese, G.L. Israel et al.: 2021. *The new magnetar SGR J1830-0645 in outburst*. ApJ Lett. 907, L34 ([arXiv:2011.08653](#),[DOI](#)).
2. **C. Dehman**, D. Viganò, N. Rea et al.: 2020. *On the rate of crustal failures in young magnetars*. ApJ Lett. 902, L32 ([arXiv:2010.00617](#),[DOI](#)).
1. P. Esposito, N. Rea, A. Borghese et al.: 2020. *A very young radio-loud magnetar*. ApJ Lett. 896, L30 ([arXiv:2004.04083](#),[DOI](#)).

### **MONOGRAPH: PhD THESIS**

**C. Dehman**: 2023. *Unveiling the Physics of Neutron Stars: A 3D expedition into MAgneto-Thermal evolution in Isolated Neutron Stars with MATINS*, Universitat Autònoma de Barcelona, Catalan Open Research Area ([arXiv:2405.00133](#),[DOI](#)).